

IT1X24 Part 2 – Tan I-Je

1. Content Knowledge Gained

Through this assignment, I gained a clearer understanding of how hypothesis testing works in real-life situations. Before this, I mainly memorised formulas for one-sample and paired t-tests, but I did not fully understand why we used them or how to interpret the results properly.

For the one-sample t-test, I learned how to compare a sample mean to a benchmark value (7 hours of sleep) and determine whether the difference is statistically significant. I better understood how the sample size, standard deviation, and difference from the benchmark affect the t-value. Seeing our test statistic fall into the rejection region made the concept much clearer.

For the paired t-test, I learned why we calculate the difference between "before" and "after" for the same individuals instead of treating them as separate groups. This helped me understand dependent samples better. I also became more confident in identifying whether a test should be left-tailed or right-tailed.

I mainly learned by revising lecture notes, practicing the formulas step-by-step, and discussing with my group members whenever I was unsure.

2. My Contribution and Teamwork

My main contribution was handling the statistical calculations and writing the conclusion statements. I wrote out the formulas clearly, substituted the values, calculated the t-statistics, and compared them with the critical values. I also ensured that our rejection rules matched the correct tail direction for each test.

Working in a group taught me the importance of dividing roles based on strengths. Some members focused more on writing and structuring the report, while others helped with data collection. We also checked each other's work to reduce mistakes. This made our report more accurate and organised.

3. Evaluation, Limitations and Improvements

Overall, I think our group selected the correct tests and explained our results clearly. The structure of our report followed the required format, and our interpretation linked back to the research statements.

However, there were limitations. We used convenience sampling, so the results may not represent all polytechnic students. The sleep data was self-reported, which may not be fully accurate. We also did not control other factors like stress or caffeine intake that could affect sleep.

To improve, future research could use a larger and more random sample, track sleep using devices instead of self-reporting, and control for more variables. This would increase the reliability and validity of the findings.

Overall, this assignment helped me see how statistics can be applied to real social issues instead of just being numbers and formulas. It improved my understanding of hypothesis testing and strengthened my ability to interpret statistical results more confidently.